Section 14.3: Partial Derivatives

Partial Derivatives

Partial Derivatives — Motivation and Definition

In Calc I, $f'(x)$ measures how f changes as x changes.

For $f(x, y)$, we have **two** input variables. How do we measure the rate of change?

Idea: Change one variable at a time, holding the other constant.

Partial Derivatives

$$f_x(x, y) = \lim_{h \to 0} \frac{f(x+h,\, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{k \to 0} \frac{f(x,\, y+k) - f(x, y)}{k}$$

Notation: All of these mean the same thing:

$$f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = D_x f$$

These are just Calc I derivatives in disguise. The only new idea: **hold the other variable constant**.

Computing Partial Derivatives

The Rule: To find f_x, treat y as a constant and differentiate with respect to x using all your Calc I rules.

For f_y, treat x as a constant and differentiate with respect to y.

You already know how to do this! Every derivative rule from Calc I (power, product, quotient, chain) still applies — you're just ignoring one variable.

Example 1: Polynomial Partials

Example 1

Find f_x and f_y for $f(x, y) = 3x^2y + xy^2 + y - 1$.

For f_x, treat y as a constant. For f_y, treat x as a constant.

Your Turn: Compute Partials

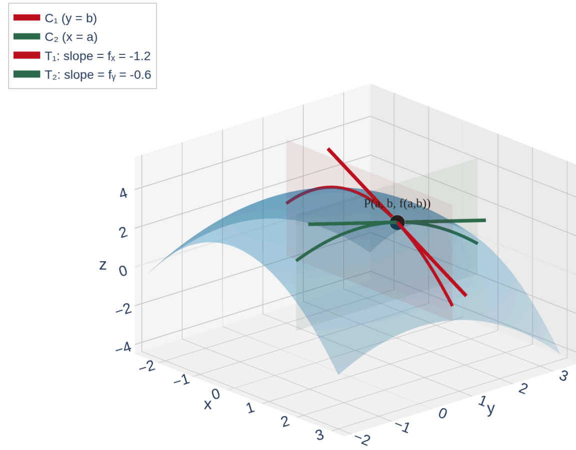
Example 2

Find f_x and f_y for $f(x, y) = x^2 \ln y + y \sin x$.

For f_x, treat y as a constant — which terms involve x? For f_y, treat x as a constant.

Geometric Interpretation of Partial Derivatives

Slice the surface $z = f(x, y)$ with the plane $y = b$.
This gives a **trace curve** C_1 in the xz-plane.
$f_x(a, b)$ = **slope of the tangent** to C_1 at the point $(a,\ b,\ f(a, b))$.
Similarly, slicing with $x = a$ gives a trace C_2, and $f_y(a, b)$ is the slope of the tangent to C_2.



Desmos 3D [https://www.desmos.com/3d]

$f_x(a, b)$ = slope in the x**-direction**. $f_y(a, b)$ = slope in the y**-direction**. Each partial derivative captures only **one direction** of change.

Example 3: Different Rules for Different Variables

Example 3

Find both partial derivatives of $f(x, y) = x^y$.

This is interesting because x and y play **different roles** — one is the base, one is the exponent.

For f_x, treat y as constant (power rule). For f_y, treat x as constant (exponential rule).

Example 4: Rates of Change — Heat Index

Example 4

The heat index I is the perceived temperature as a function of actual temperature T and relative humidity H. Use the table to estimate $f_T(96, 70)$ and $f_H(96, 70)$, and interpret each.

$T \setminus H$	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	**125**	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

Heat index values for temperature T (°F) and relative humidity H (%)

Use the difference quotient with nearby table values. Hold one variable constant while varying the other.

Higher-Order Partial Derivatives

Just like in Calc I, we can differentiate **again** to get second (and higher) derivatives.

For $f(x, y)$, there are **four** second-order partial derivatives:

- $f_{xx} = \dfrac{\partial^2 f}{\partial x^2}$ — differentiate twice with respect to x
- $f_{yy} = \dfrac{\partial^2 f}{\partial y^2}$ — differentiate twice with respect to y
- $f_{xy} = \dfrac{\partial^2 f}{\partial y\, \partial x}$ — first x, then y
- $f_{yx} = \dfrac{\partial^2 f}{\partial x\, \partial y}$ — first y, then x

Interpretation:

- f_{xx} = concavity in the x-direction (is the surface curving up or down as you walk in x?)
- f_{yy} = concavity in the y-direction
- f_{xy} = how the x-slope changes as you move in y (a **mixed** partial)

Notation warning: f_{xy} means differentiate with respect to x **first**, then y. Read the subscripts left to right.

Clairaut's Theorem

Are f_{xy} and f_{yx} always equal? That is, does the **order of differentiation** matter?

Theorem: Clairaut's Theorem

Suppose f is defined on a disk D containing the point (a, b).
If f_{xy} and f_{yx} are both **continuous** on D, then:

$$f_{xy}(a, b) = f_{yx}(a, b)$$

Practical use: For polynomials, trig functions, exponentials, and all the "nice" functions we encounter in this course, mixed partials are equal:

$$f_{xy} = f_{yx}, \qquad f_{xxy} = f_{xyx} = f_{yxx}, \qquad \text{etc.}$$

For any function built from standard Calc I functions, the **order of differentiation doesn't matter**. Differentiate in whichever order is easiest!

The Clairaut Puzzle

Example 5

You are told that there is a function f whose partial derivatives are:

$$f_x(x, y) = x + 4y \qquad \text{and} \qquad f_y(x, y) = 3x - y$$

Should you believe it?

Check whether Clairaut's theorem is satisfied.

Example 6: Laplace's Equation

Example 6

Verify that $u = \ln \sqrt{x^2 + y^2}$ satisfies **Laplace's equation**:

$$u_{xx} + u_{yy} = 0$$

Laplace's equation appears throughout physics — it describes steady-state heat flow, electrostatics, and fluid flow.

Simplify u first, then find u_x, u_xx, u_y, u_yy, and add the second derivatives.

Example 7: Implicit Partial Differentiation

Example 7

If $yz = \ln(x + z)$, find $\dfrac{\partial z}{\partial x}$.

When z is defined **implicitly** as a function of x and y, we differentiate both sides with respect to x, treating y as a constant and z as a function of x.

Differentiate both sides with respect to x. Remember z depends on x, so use the chain rule.

Homework

Section 14.3: 1, 2, 4, 5, 6, 9-36 odd, 39, 41, 45, 51, 63, 64, 80, 84a, 85